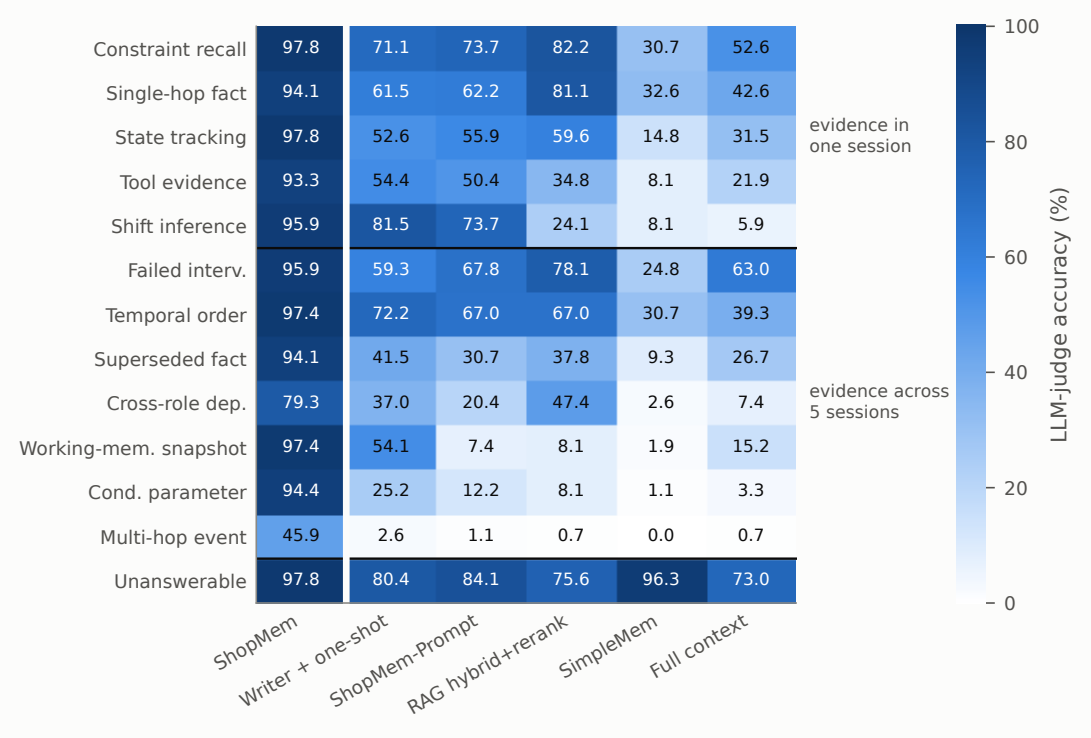
Constraint recall
Single-hop fact
State tracking
Tool evidence
Shift inference
Failed interv.
Temporal order
Superseded fact
Cross-role dep.
Working-mem. snapshot
Cond. parameter
Multi-hop event
Unanswerable
97.8 71.1 73.7 82.2 30.7 52.6
94.1 61.5 62.2 81.1 32.6 42.6
97.8 52.6 55.9 59.6 14.8 31.5
93.3 54.4 50.4 34.8 8.1 21.9
95.9 81.5 73.7 24.1 8.1 5.9
95.9 59.3 67.8 78.1 24.8 63.0
97.4 72.2 67.0 67.0 30.7 39.3
94.1 41.5 30.7 37.8 9.3 26.7
79.3 37.0 20.4 47.4 2.6 7.4
97.4 54.1 7.4 8.1 1.9 15.2
94.4 25.2 12.2 8.1 1.1 3.3
45.9 2.6 1.1 0.7 0.0 0.7
97.8 80.4 84.1 75.6 96.3 73.0
evidence in one session
evidence across 5 sessions
ShopMem
Writer + one-shot
ShopMem-Prompt
RAG hybrid+rerank
SimpleMem
Full context
LLM-judge accuracy (%)
100
80
60
40
20
0